

# Clinical Digital Twin Report

## Computational Cellular Reprogramming Analysis

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Total Cells: 2000 | Model Mode: CLINICAL

### 1. Introduction

Cellular reprogramming is an essential technique in regenerative medicine, offering potential for generating induced pluripotent stem cells (iPSCs) from somatic cells. The objective of this experiment was to investigate the effects of repeated OSKM transfection and oncogenic stress on cellular fate determination, with a particular focus on the formation of iPSCs and other cell lineages.

### 2. Materials and Methods

In this experiment, somatic cells were subjected to repeated transfection with the Yamanaka factors (Oct4, Sox2, Klf4, and c-Myc), a process known as OSKM transfection, conducted six times consecutively. Additionally, oncogenic stress was induced to assess its impact on cellular reprogramming, though no single-cell variability inference (scVI) analyses were performed.

#### Experimental Interventions:

- Oncogenic Stress Induced
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

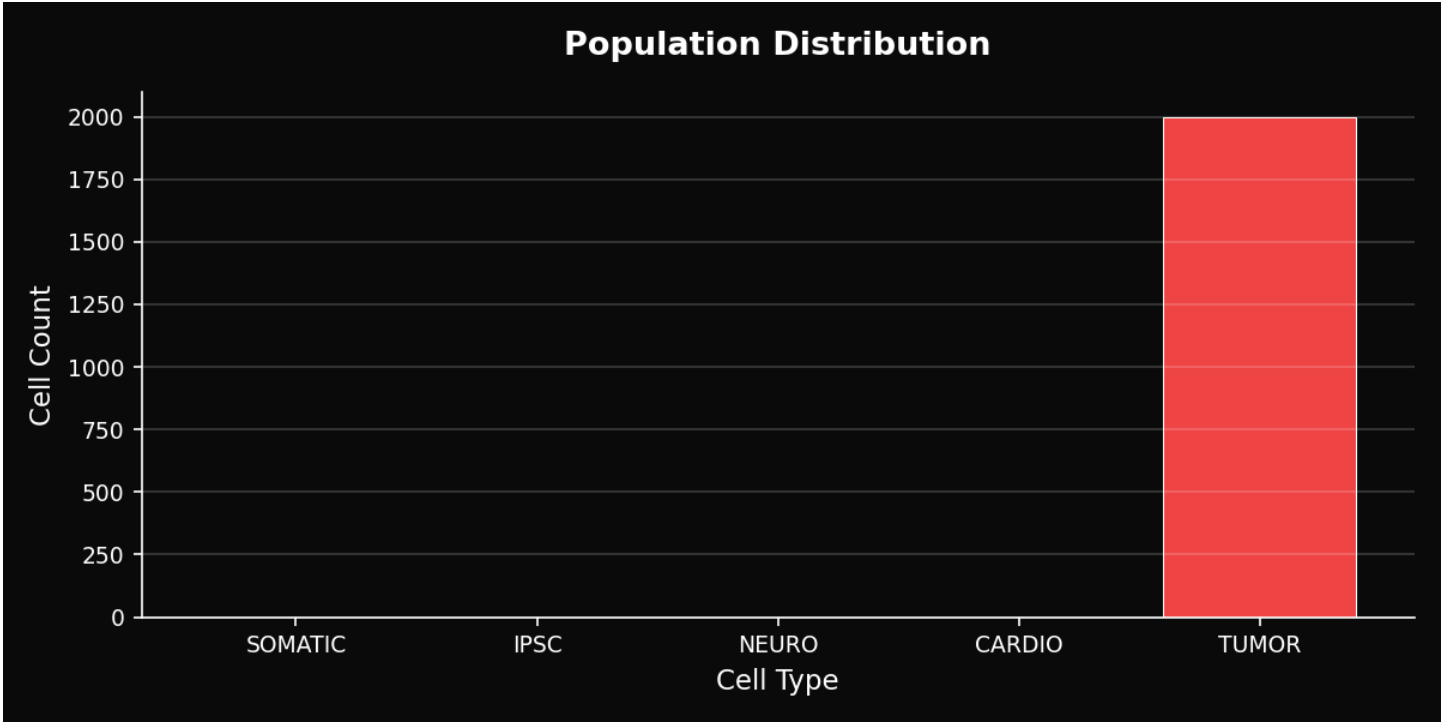
### 3. Results

The experimental findings revealed a complete lack of somatic, iPSC, neurogenic, and cardiogenic cell populations, while a significant tumorigenic population was observed, totaling 2000 cells. This indicates a predominant shift towards tumor cell formation, suggesting that the interventions were suboptimal for iPSC generation under the given conditions.

#### 3.1 Population Distribution

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| Cell Type | Count | Percentage |
|-----------|-------|------------|
| SOMATIC   | 0     | 0.0%       |
| IPSC      | 0     | 0.0%       |
| NEURO     | 0     | 0.0%       |
| CARDIO    | 0     | 0.0%       |
| TUMOR     | 2000  | 100.0%     |

## 4. Discussion

The absence of iPSCs and other differentiated cell types, coupled with the exclusive emergence of 2000 tumor cells, underscores a critical imbalance in the reprogramming protocol. The repeated OSKM transfections, in conjunction with oncogenic stress, likely facilitated oncogenic transformation instead of desired pluripotency. These findings highlight the necessity for careful modulation of reprogramming factors and stress conditions to prevent tumorigenesis and achieve successful cellular reprogramming.

*This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.*